



Improving Metadata and Discoverability

Making your software findable

Tim Dennis
UCLA OSPO

2026-09-23

Findability comes from description

A Zenodo record with a bare title and no keywords is a catalog record with no subject headings: retrievable if you already know it exists, invisible if you don't.

The tools are GitHub topics, Zenodo fields, and a README. The question is the same: *what would a seeker search for, and does this record contain it?*

You already have a license, a CITATION.cff, and a DOI. **This episode ties them together.**

What counts as useful metadata?

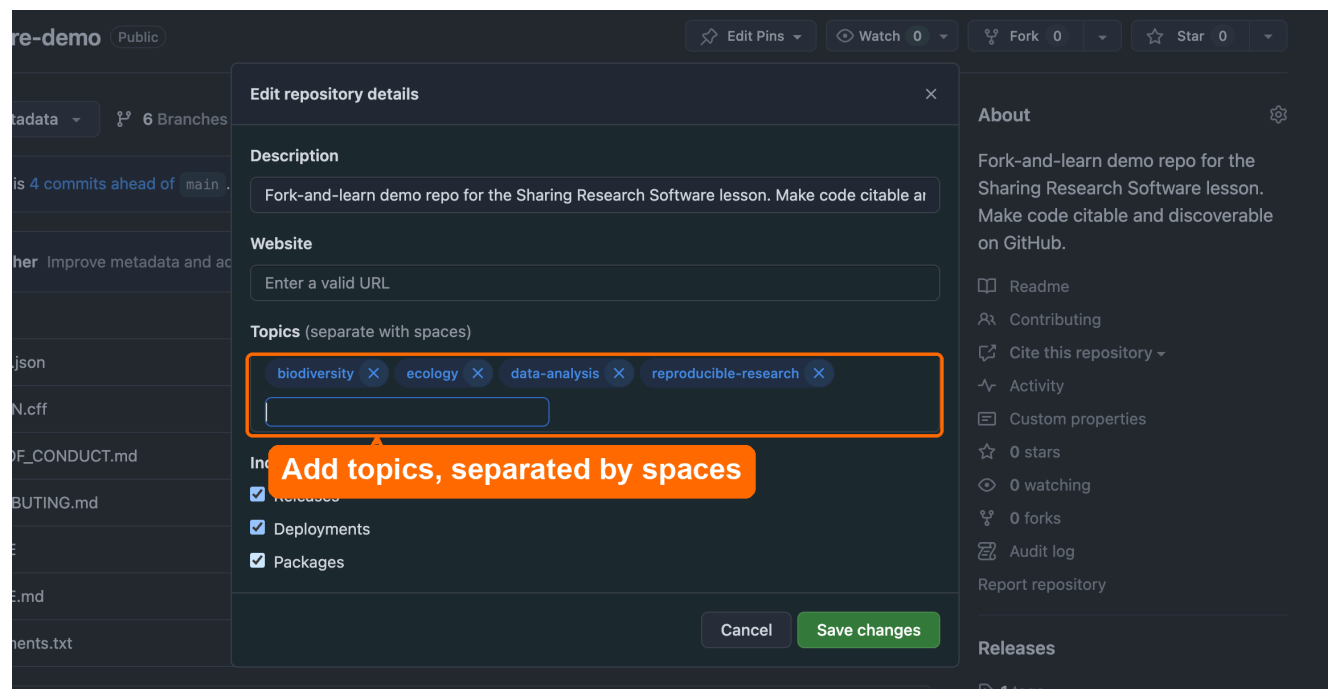
Good metadata answers predictable questions with minimal effort from the reader:

- **What is this software?** A short description or abstract
- **Who made it?** Authors and ORCIDs
- **How do I cite it?** CITATION.cff and a DOI
- **What field is it for?** Keywords
- **What else does it relate to?** A related article, dataset, or funder

Add GitHub Topics

On your repo's front page, click the **gear icon next to About**, then fill in **Topics**:

- **Discipline:** `geospatial`, `text-mining`, `materials-science`
- **Method:** `simulation`, `visualization`, `machine-learning`
- **Language:** `python`, `r`



The 30-second rule

Your README is your software's front door.

If people can't tell what it does, how to install it, and how to use it in **30 seconds**, they leave.

A repo without a real README is a collection without a finding aid: only its author can use it.

README: 7 essential sections

1. **About:** what it does, in 2-3 sentences
2. **Features:** key capabilities
3. **Getting Started:** prerequisites and install
4. **Usage:** a minimal working example
5. **Citation:** link to CITATION.cff or the DOI
6. **License:** state the terms
7. **Contact:** how to get help

[UC OSPO README Guide](#) (*UC-specific*)

Challenge: give your README the 7 sections

 5 min

In your fork of `software-demo` :

1. Open `README.md` and click the **pencil icon** to edit
2. Add a `##` heading for each of the 7 sections: About, Features, Getting Started, Usage, Citation, License, Contact
3. Fill in what you can, then **Commit changes** to `main`

Short on time? Keep your title and DOI badge, **replace everything below them** with the starter README from the Etherpad, then replace `YOUR-USERNAME` , `YOUR-DOI` , and `YOUR-RECORD-ID` (from your Sandbox record's web address).

Next: compare yours with the reference README.

Before → after

✗ Before (`main` , starting state)

```
1 # Biodiversity Analysis Toolkit
2
3 Analysis tools for biodiversity
4 research data.
```

One line of description. No install steps, no usage, no citation, no license, no one to contact.

✓ After (`05-add-metadata` , abridged)

```
1 # Biodiversity Analysis Toolkit
2 [DOI badge] [License badge]
3 Analysis tools for biodiversity... <!-- 1 About -->
4
5 ## Features <!-- 2 -->
6 - Citable, open source, documented
7
8 ## Getting Started <!-- 3 -->
9 Prerequisites: Python 3.9+
10 ### Installation & Usage <!-- 4 -->
11 pip install -r requirements.txt
12 python src/analysis.py
13
14 ## Citation <!-- 5 -->
15 Cite via CITATION.cff or the DOI badge.
16 ## License <!-- 6 -->
17 BSD 3-Clause. See LICENSE.
18 ## Contact <!-- 7 -->
19 Open an issue in this repository.
```

Beyond the README: community health files

CONTRIBUTING.md

How to contribute. Add one *if* you want contributions, or say plainly that you don't.

CODE_OF_CONDUCT.md

Behavioral standards. [Contributor Covenant](#) is the standard choice.

CHANGELOG.md

Version history: what's new, what broke, what changed.

GitHub recognizes certain filenames

File	What GitHub does
README.md	Renders as the repository's landing page
LICENSE	Shows the license in the repo sidebar
CITATION.cff	Adds "Cite this repository" with APA/BibTeX
CODE_OF_CONDUCT.md	Counts toward the community profile
CONTRIBUTING.md	Links guidelines when someone opens an issue/PR
SECURITY.md	Becomes the security policy

These only appear once the file is on the default branch. GitHub changes the details, so teach the pattern: the filename is the signal.

Where should software live?

Zenodo is the recommended deposit location for software:

- Archives each GitHub release automatically
- Mints a DOI for every version
- Registers records with DataCite, which discovery services harvest
- Free, run by CERN, recognized by journals and funders

Dataverse and Dryad are good for **datasets**, but don't auto-integrate with GitHub releases the same way.

UC-specific: all UC campuses have Dryad. If you deposit data there, Dryad prompts a linked Zenodo code deposit automatically.

Zenodo metadata is what discovery systems see

Zenodo registers each record with **DataCite**. Search tools and discovery services pick it up from there, so the Zenodo record is what they show.

Add or refine: authors + ORCIDs, keywords, related works, funding, a readable description.

Caution: if a repo has a `.zenodo.json`, Zenodo uses *that* file, not `CITATION.cff`. Keep them in sync or you'll publish two different citation records for the same software.

Supporting others: priority order when metadata is thin

1. A readable **description**, the biggest loss when it's missing
2. **Author ORCIDs**: without them, every J. Chen on Zenodo looks the same
3. **Keywords**, reused consistently across GitHub, CITATION.cff, Zenodo
4. **Related works**: link the software to its paper, dataset, and grant

Challenge: improve your Zenodo record

 5 min

Open your record on **Zenodo Sandbox** (sandbox.zenodo.org) from the release episode. Select **Edit**, and add:

- keywords
- author ORCIDs, if they're missing
- a real description
- *optional*: a related paper or funder, if you have one

Save and publish the updated record when done.

Challenge: which change makes it discoverable?

 3 min

A tool has been public on GitHub for a year: one-line README, no topics, a Zenodo record with only a title. **Which single change matters most?**

- A. Add collaborators
- B. Descriptive topics, matching keywords, and a description
- C. More frequent commits
- D. A shorter name

B. People find software through its *description*. Commit activity doesn't help. It's the same reason a catalog record with no subject headings is nearly invisible.

Key points

- Description is what makes software findable, on GitHub, Zenodo, and beyond
- Keep it consistent across CITATION.cff, README, topics, and Zenodo: that's authority control

Get involved

Join

- **Slack:** day-to-day questions and resources
- **Newsletter:** sign up from the Contact page
- **Your campus mailing list:** also on the Contact page

Show up

- **All-campus virtual meetup:** first Thursday of each month, 10am
- **Coworking & office hours:** Tuesdays 10am, Thursdays 1pm
- **UCSB Open Source Lounge:** Thursdays 2:30-4:30pm, hybrid, open to all UC

Start here: ucospo.net/about/get-involved · Questions: Laura Langdon, Community Manager

All times Pacific. Zoom links and calendar feeds: ucospo.net/events